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LIQUID JET HEAD AND LIQUID JET RECORDING APPARATUS

Abstract

Visual confirmation of frame ground coupling at the time of assembling is made easy. A liquid jet head according to an aspect of the present disclosure is provided with a jet module configured to jet a liquid, and a coupling member having conductivity and configured to couple the jet module to the frame ground, wherein at least a part of the coupling member is disposed outside the jet module.

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Background/Summary

RELATED APPLICATIONS

[0001] This application claims priority to Japanese Patent application Nos. JP 2024-026460, filed on Feb. 26, 2024, the entire content of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] An embodiment of the present disclosure relates to a liquid jet head and a liquid jet recording apparatus.

2. Description of the Related Art

[0003] JP2006-68981 A discloses a liquid jet head including a head unit including a flow path unit provided with a liquid flow path, a nozzle plate having conductivity and bonded to the head unit, a head case to which the nozzle plate and the head unit are fixed, a head cover having conductivity and attached to the head case so as to enclose the nozzle plate and the head unit from the outside. The head cover includes a frame part provided with an opening part for exposing the nozzle plate, and a contact protrusion protruding inward from an inner circumferential edge of the frame part. The head cover is attached to the head case in a state in which the contact protrusion has contact with an outer circumferential edge of the nozzle plate.

[0004] JP6504889B (PTL2) discloses a configuration including a main body, an ejection unit capable of ejecting a liquid supplied through a liquid flow path, an electrical wiring board provided with a contact point part for receiving a signal from the outside, a flow path formation member which forms the liquid flow path, and a plate spring having conductivity and electrically coupled to the contact point part. In PTL2, contact pressure necessary for the electrical coupling between the plate spring and the contact point part is ensured by an elastic deformation of the plate spring.

[0005] However, in the case of PTL1, since the contact protrusion is disposed inside the head cover, it is difficult to visually confirm whether the contact protrusion has contact with the nozzle plate when viewed from the outside of the head cover. In the case of the configuration in PTL2, it is difficult to visually confirm whether the plate spring and the contact point part have coupling with each other when viewed from the outside of the main body. Further, when the whole of a coupling member for coupling the main body to a frame ground is disposed inside the main body, it becomes difficult to visually confirm the frame ground coupling at the time of assembling.

[0006] The present disclosure is made in view of the problem described above, and has an object of making it easy to visually confirm the frame ground coupling at the time of assembling.

SUMMARY OF THE INVENTION

[0007] (1) A liquid jet head according to an aspect of the present disclosure is provided with a jet module configured to jet a liquid, and a coupling member having conductivity and configured to couple the jet module to the frame ground, wherein at least a part of the coupling member is disposed outside the jet module.

[0008] For example, when the whole of the coupling member for coupling the jet module to the frame ground (hereinafter also referred to as “FG”) is disposed inside the jet module, it becomes difficult to visually confirm the FG coupling at the time of assembling. In contrast, according to the liquid jet head related to the present aspect, since at least a part of the coupling member is disposed outside the jet module, it is easy to visually confirm the FG coupling at the time of assembling. Further, it is easy to confirm the conduction even after assembling.

[0009] (2) In the liquid jet head according to the aspect (1), the coupling member may be fixed outside the jet module.

[0010] According to this configuration, the fixation work can be performed outside the jet module, which, therefore, makes a contribution to an improvement of the workability.

[0011] (3) In the liquid jet head according to the aspect (2), the coupling member may be fixed on a side surface in a longitudinal direction of the jet module.

[0012] According to this configuration, since there is no influence in the thickness direction of the liquid jet head, density growth can be achieved even when disposing a plurality of liquid jet heads in the thickness direction.

[0013] (4) In the liquid jet head according to any one of the aspects (1) to (3), the coupling member

may be fixed with a screw.
[0014] According to this configuration, since screw fixation is adopted, the conduction can more reliably be achieved.
[0015] (5) In the liquid jet head according to any one of the aspects (1) to (4), there may further be included a stay disposed outside the jet module, wherein a part of the coupling member may protrude outside the stay.
[0016] According to this configuration, since a part of the coupling member protrudes outside the stay, it is easy to visually confirm the FG coupling at the time of assembling.
[0017] (6) In the liquid jet head according to the aspect (5), there may further be included a holding portion configured to clamp the part of the coupling member protruding outside the stay with the stay to hold the part.
[0018] According to this configuration, the fixation is achieved by the clamping with the stay and the holding portion, which, therefore, makes a contribution to an improvement in reliability of the electrical coupling.
[0019] (7) In the liquid jet head according to one of the aspects (5) and (6), the stay may be provided with an opening part through which the coupling member passes.
[0020] According to this configuration, it is possible to draw a part of the coupling member out of the stay through the opening part of the stay.
[0021] (8) In the liquid jet head according to one of the aspects (6) and (7), there may further be included a cover configured to cover the jet module and the stay, wherein the holding portion may form a part of the cover.
[0022] According to this configuration, since the fixation is achieved by the clamping with a part of the cover and the stay, no dedicated fixation component is required, which makes a contribution to a reduction in the number of components.
[0023] (9) In the liquid jet head according to any one of the aspects (6) to (8), there may further be included a plate member having the holding portion.
[0024] According to this configuration, it is possible to more reliably achieve the conduction compared to when fixing the coupling members directly with the screw.
[0025] (10) A liquid jet recording apparatus according to an aspect of the present disclosure includes the liquid jet head according to any one of the aspects (1) to (9).
[0026] According to the liquid jet recording apparatus related to the present aspect, it is possible to obtain the liquid jet recording apparatus which is easy to visually confirm the FG coupling at the time of assembling, and is easy to confirm the conduction even after assembling.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0027] FIG. **1** is a schematic configuration diagram of an inkjet printer according to a first embodiment.
[0028] FIG. **2** is a perspective view of an inkjet head according to the first embodiment.
[0029] FIG. **3** is a partially exploded perspective view of the inkjet head according to the first embodiment.
[0030] FIG. **4** is a perspective view of the inkjet head according to the first embodiment with a part thereof (an FPC unit) exposed.
[0031] FIG. **5** is a diagram viewed from the arrow V shown in FIG. **2**.
[0032] FIG. **6** is a perspective view of a nozzle guard related to the first embodiment.
[0033] FIG. **7** is an enlarged view of the boxed portion VII in FIG. **2**.
[0034] FIG. **8** is a diagram illustrating a conduction path via a base member related to the first embodiment.

[0035] FIG. **9** is a diagram illustrating a conduction path sandwiched by a cover and a stay related to the first embodiment.
[0036] FIG. **10** is a partially exploded perspective view of an inkjet head according to a second embodiment.
[0037] FIG. **11** is a perspective view showing an attachment portion of a stay and a plate member related to the second embodiment.
[0038] FIG. **12** is a diagram illustrating a conduction path sandwiched by the plate member and the stay related to the second embodiment.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0039] Some embodiments according to the present disclosure will hereinafter be described with reference to the drawings. In the embodiments and modified examples hereinafter described, constituents corresponding to each other will be denoted by the same reference symbols to omit the descriptions thereof in some cases. In the following descriptions, expressions representing relative or absolute arrangements such as "parallel," "perpendicular," "central," and "coaxial" not only represent strictly such arrangements, but also represent the state of being relatively displaced with a tolerance, or an angle or a distance to the extent that the same function can be obtained.
[0040] In the following embodiments, the description will be presented citing an inkjet printer (hereinafter referred to simply as a "printer") for performing recording on a recording target medium using ink (a liquid) as an example of the liquid jet recording apparatus equipped with the liquid jet head according to the present disclosure. The scale size of each member is arbitrarily modified so as to provide a recognizable size to the member in the drawings used in the following description.

First Embodiment

Printer

[0041] FIG. **1** is a schematic configuration diagram of a printer **1**.
[0042] As shown in FIG. **1**, the printer **1** according to the present embodiment is provided with a pair of conveyance mechanisms **2**, **3**, an ink supply mechanism **4**, inkjet heads **5**A, **5**B (liquid jet heads), and a scanning mechanism **6**.
[0043] In the following explanation, the description is presented using an orthogonal coordinate system of X, Y, and Z as needed. The X direction corresponds to a conveyance direction of recording target medium P (an example of a medium such as paper). The Y direction corresponds to the scanning direction of the scanning mechanism **6**. The Z direction corresponds to a vertical direction perpendicular to the X direction and the Y direction. In the following explanation, the description will be presented defining the arrow direction as the positive (+) direction, and a direction opposite to the arrow direction as the negative (−) direction in the drawings in each of the X direction, Y direction, and the Z direction. In the present embodiment, the +Z direction corresponds to an upward direction in the gravitational direction, and the −Z direction corresponds to a downward direction in the gravitational direction.
[0044] The conveyance mechanisms **2**, **3** convey the recording target medium P in the X direction. Specifically, the conveyance mechanism **2** is provided with a grit roller **11** extending in the Y direction, a pinch roller **12** extending in parallel to the grit roller **11**, and a drive mechanism (not shown) such as a motor for making the grit roller **11** make an axial rotation. Similarly, the conveyance mechanism **3** is provided with a grit roller **13** extending in the Y direction, a pinch roller **14** extending in parallel to the grit roller **13**, and a drive mechanism (not shown) for making the grit roller **13** make an axial rotation.
[0045] The ink supply mechanism **4** is provided with ink tanks **15** each housing the ink, and ink pipes **16** for respectively coupling the ink tanks **15** and the inkjet heads **5**A, **5**B to each other.
[0046] In the present embodiment, the ink tanks **15** are arranged in the X direction. The ink tanks **15** respectively house four colors of ink such as yellow ink, magenta ink, cyan ink, and black ink.
[0047] The ink pipes **16** are each, for example, a flexible hose having flexibility. The ink pipes **16**

connect the ink tanks **15** and the inkjet heads **5**A, **5**B to each other, respectively.
[0048] The scanning mechanism **6** reciprocates the inkjet heads **5**A, **5**B in the Y direction. Specifically, the scanning mechanism **6** is provided with a pair of guide rails **21**, **22**, a carriage **23**, and a drive mechanism **24**, wherein the pair of guide rails **21**, **22** extend in the Y direction, the carriage **23** is movably supported by the pair of guide rails **21**, **22**, and the drive mechanism **24** moves the carriage **23** in the Y direction.
[0049] The drive mechanism **24** is disposed between the guide rails **21**, **22** in the X direction. The drive mechanism **24** is provided with a pair of pulleys **25**, **26**, an endless belt **27**, and a drive motor **28**, wherein the pair of pulleys **25**, **26** are disposed at an interval in the Y direction, the endless belt **27** is wound between the pair of pulleys **25**, **26**, and the drive motor **28** rotationally drives the pulley **26** as one of the pulleys **25**, **26**.
[0050] The carriage **23** is coupled to the endless belt **27**. On the carriage **23**, there is mounted the plurality of inkjet heads **5**A, **5**B in the state of being arranged side by side in the Y direction. The inkjet heads **5**A, **5**B are arranged so that the two colors of ink can be ejected from each of the inkjet heads **5**A, **5**B. Therefore, in the printer **1** according to the present embodiment, there is adopted the configuration in which the inkjet heads **5**A, **5**B each eject the two colors of ink, wherein the two colors of ink ejected by the inkjet head **5**A are different from the two colors of ink ejected by the inkjet head **5**B, and thus, the four colors of ink, namely the yellow ink, the magenta ink, the cyan ink, and the black ink, can be ejected.
Inkjet Head
[0051] FIG. **2** is a perspective view of the inkjet head **5**A. FIG. **3** is a partially exploded perspective view of the inkjet head **5**A. FIG. **4** is a perspective view of the inkjet head **5**A with a part (an FPC unit **100**) thereof exposed. FIG. **5** is a diagram viewed from the arrow V shown in FIG. **2**. FIG. **6** is a perspective view of a nozzle guard **40**. It should be noted that the inkjet heads **5**A, **5**B have equivalent configurations except the colors of the ink supplied. Therefore, in the following explanation, the inkjet head **5**A will be described, and the description of the inkjet head **5**B will be omitted.
[0052] The inkjet head **5**A according to the present embodiment is an inkjet head of an electromechanical transduction system in which the ink is ejected from a head chip including an actuator plate formed of a piezoelectric element made of PZT (lead zirconate titanate) or the like.
[0053] In this inkjet head **5**A, in order to eject the ink, a voltage is applied between electrodes on drive walls of an ejection channel provided to the actuator plate to cause the drive wall to make a thickness-shear deformation. Thus, due to a change in volume of the ejection channel, the ink in the ejection channel is ejected through a nozzle hole. It should be noted that an ejection system of the ink is not limited to the electromechanical transduction system described above, and it is possible to adopt a charge control system, a pressure vibration system, an electrothermal transduction system, an electrostatic suction system, and so on.
[0054] The charge control system is for providing a charge to a material with a charge electrode to eject the material from a nozzle while controlling a flight direction of the material with a deflection electrode. Further, the pressure vibration system is for applying super high pressure to a material to eject the material toward a nozzle tip, and when a control voltage is not applied, the material goes straight to be ejected from the nozzle, and when the control voltage is applied, an electrostatic repelling force is generated between the materials, and the material flies in all directions to be prevented from being ejected from the nozzle.
[0055] Further, the electrothermal transduction system is for rapidly vaporizing a material with a heater provided in a space retaining the material to generate a bubble, to eject the material located in the space with the pressure of the bubble. The electrostatic suction system is for applying minute pressure to a space retaining a material to form a meniscus in the nozzle, applying an electrostatic attractive force in this state, and then pulling the material out. Further, besides the above, it is possible to adopt technologies such as a system using a viscosity alteration of a fluid due to an

electric field, or a system of flying a material with a discharge spark.
[0056] Referring to FIG. **2** to FIG. **6** in combination, the inkjet head **5**A according to the present embodiment is configured with jet modules **30**A, **30**B, and **30**C (see FIG. **3**), a nozzle plate **35** (see FIG. **5**), the nozzle guard **40** (see FIG. **6**), and so on mounted on a base member **50**.

Base Member

[0057] The base member **50** is formed to have a shape having a thickness direction set to the Z direction, and a longitudinal direction set to the X direction. The base member **50** includes a base main body part **51** for holding the jet modules **30**A, **30**B, and **30**C, and a carriage fixation part **52** for fixing the base member **50** to the carriage **23** (see FIG. **1**). In the present embodiment, the base member **50** is formed as a separated body from a frame body **36**. For example, the base member **50** may be formed of a metal material such as stainless steel.
[0058] The base main body part **51** is provided with a module housing part **53** which opens in the Z direction so as to be able to house the jet modules **30**A, **30**B, and **30**C. It is arranged that the jet modules **30**A, **30**B, and **30**C can be inserted into the module housing part **53**. A −Z-direction end portion of each of the jet modules **30**A, **30**B, and **30**C is arranged to be able to be inserted into the module housing part **53** from the +Z direction side. In this insertion state, the jet modules **30**A, **30**B, and **30**C are each held by the base main body part **51** in a state of standing up toward the +Z direction from the base member **50**.
[0059] The carriage fixation part **52** projects from a +Z-direction end portion of the base main body part **51** in the X-Y plane. The carriage fixation part **52** more largely projects outward in the X direction than in the Y direction. The carriage fixation part **52** is provided with attachment holes or the like for attaching the base member **50** to the carriage **23** (see FIG. **1**).

Jet Modules

[0060] The jet modules **30**A, **30**B, and **30**C are formed to have a plate-like shape having a thickness direction set to the Y direction, and a longitudinal direction set to the X direction. The jet modules **30**A, **30**B, and **30**C are each configured so as to be able to eject the ink supplied from the ink tank **15** (see FIG. **1**) toward the recording target medium P. The jet modules **30**A, **30**B, and **30**C are mounted on the base member **50** in a state of being arranged side by side in the Y direction.
[0061] In the inkjet head **5**A according to the present embodiment, it is arranged that the jet modules **30**A, **30**B, and **30**C are grouped into a single port, and therefore eject ink of a single color. It should be noted that the number of the jet modules **30**A, **30**B, and **30**C mounted on the base member **50**, and a color, a type, and so on of the ink to be ejected from the jet modules **30**A, **30**B, and **30**C can be changed as appropriate. The jet modules **30**A, **30**B, and **30**C are each formed of the same configuration, and are mounted on the base member **50** in a state in which the jet modules **30**A, **30**B, and **30**C are arranged side by side in the Y direction.
[0062] In the present embodiment, each of the jet modules **30**A, **30**B, and **30**C is of a so-called edge-shoot type which ejects the ink from an end portion (e.g., a −Z-direction end surface of the head chip) in the extending direction (the Z direction) in an ejection channel.
[0063] In each of the jet modules **30**A, **30**B, and **30**C, the FPC unit **100** is supported by a surface facing to the −Y direction. The FPC unit **100** is provided with a drive board **101** and a wiring board **102**. For example, the drive board **101** and the wiring board **102** are each a flexible printed board, and are each formed of a base film provided with wiring patterns formed thereon. It should be noted that the drive board **101** may be formed of a rigid board or the like in a portion (a mounting portion) on which a driver for driving the head chip is mounted.
[0064] The base member **50** is provided with a stay **60** for supporting a component mounted on the base member **50**. The stay **60** stands up toward the +Z direction from the base member **50**, and at the same time, collectively surrounds the periphery of the jet modules **30**A, **30**B, and **30**C.

Nozzle Plate

[0065] In the present embodiment, the nozzle plate **35** (see FIG. **5**) is formed of a resin material such as polyimide. The nozzle plate **35** is fixed to the-Z-direction end surface of the base main body part **51** and the −Z-direction end surfaces (portions exposed from the module housing part **53**) of the jet modules **30**A, **30**B, and **30**C via an adhesive or the like.

[0066] The nozzle plate **35** is provided with nozzle holes (not shown) penetrating the nozzle plate **35** in the Z direction. The nozzle holes are independently formed at positions opposed in the Z direction to the respective ejection channels of the head chip.

[0067] It should be noted that the nozzle plate **35** is not limited to the resin material. For example, the nozzle plate **35** may also be formed of silicon or a metal material (stainless steel or the like), and may also be provided with a layered structure of the resin material and the metal material. Further, there may be adopted a configuration in which the jet modules **30**A, **30**B, and **30**C are covered with a single nozzle plate **35** in a lump, or a configuration in which the jet modules **30**A, **30**B, and **30**C are individually covered with a plurality of nozzle plates **35**.

Nozzle Guard

[0068] The nozzle guard **40** is a metallic member for protecting the nozzle plate **35**. The nozzle guard **40** is formed by applying hybrid processing of drawing processing and bending processing to a plate member made of, for example, stainless steel. The nozzle guard **40** covers the base main body part **51** from the −Z direction side in the state of sandwiching the nozzle plate **35** in-between. It should be noted that the frame body **36** (an example of a housing made of resin) formed of synthetic resin such as plastic may be disposed between the nozzle guard **40** and the base member **50**.

[0069] In the nozzle guard **40**, at a position opposed in the Z direction to the −Z-direction end surface of each of the jet modules **30**A, **30**B, and **30**C, there is formed an exposure hole **41***h* for exposing the nozzle plate **35** to the outside. The exposure hole **41***h* is formed to have a slit-like shape penetrating the nozzle guard **40** in the Z direction, and extending in the X direction. The exposure holes **41***h* are formed in three lines at intervals in the Y direction so as to correspond to the respective jet modules **30**A, **30**B, and **30**C. The nozzle holes described above are communicated with the outside of the inkjet head **5**A through the exposure holes **41***h*.

Conductive Parts

[0070] FIG. **7** is an enlarged view of the boxed portion VII in FIG. **2**. FIG. **8** is a diagram illustrating a conduction path via the base member **50**. In FIG. **8**, the illustration of the frame body **36** disposed between the nozzle guard **40** and the base member **50** is omitted.

[0071] Referring to FIG. **7** and FIG. **8** in combination, the inkjet head **5**A is provided with the base member **50**, the jet modules **30**A, **30**B, and **30**C which are mounted on the base member **50** and jet the ink, the nozzle plate **35** provided with the nozzle holes for jetting the ink, the nozzle guard **40** made of metal for protecting the nozzle plate **35**, and conductive parts **43** which have conductivity, are electrically coupled to the jet modules **30**A, **30**B, and **30**C and the nozzle guard **40** via the base member **50**, and couple the nozzle guard **40** to FG.

[0072] The conductive parts **43** are fixed outside the jet modules **30**A, **30**B, and **30**C. Specifically, the conductive parts **43** are each fixed at an outer side in the X direction of an outer end portion in the X direction of each of the jet modules **30**A, **30**B, and **30**C.

[0073] The conductive parts **43** are fixed on a side surface in the longitudinal direction (the X direction) of each of the jet modules **30**A, **30**B, and **30**C. It should be noted that the conductive parts **43** are not fixed on a side surface in the thickness direction (the Y direction) of each of the jet modules **30**A, **30**B, and **30**C.

[0074] Referring to FIG. **6** in combination, at least corner portions of the nozzle guard **40** have drawn parts **45** to which drawing processing is applied. The nozzle guard **40** is provided with a guard main body part **41** formed to have a plate shape having the thickness direction set to the Z direction, and the longitudinal direction set to the X direction, and an uprise part **42** rising toward the +Z direction from an outer circumferential edge of the guard main body part **41**. The drawn parts **45** are disposed at four corner portions of a rectangular shape having the longitudinal direction set to the X direction when viewing the nozzle guard **40** from the Z direction.

[0075] The side surface portion of the nozzle guard **40** has a bent part **46** to which the bending processing is applied. The bent part **46** is disposed between a portion (a planar portion) of the uprise part **42** except the corner portions and the guard main body part **41**. The uprise part **42** rises toward the +Z direction from each of an outer edge in the X direction and an outer edge in the Y direction of the guard main body part **41** via the bent part **46**.
[0076] On the boundary between the drawn part **45** and the bent part **46** of the nozzle guard **40**, there is formed a cutout **47**. The cutout **47** is formed to have a size enough to discharge a liquid which has entered the drawn part **45**. The cutouts **47** are formed at four places so as to correspond to the drawn parts **45** disposed at the four corner portions. The cutouts **47** are formed at positions at the inner side in the Y direction of the respective drawn parts **45**. The cutouts **47** are each formed longer in the Z direction than in the X direction, and each open in the Y direction.
[0077] The conductive part **43** is formed of the same member as, and integrally with, the side surface portion of the nozzle guard **40**. In the nozzle guard **40**, the side surface portion including the conductive part **43** is formed to have an L-shape when viewed from the X direction. A screw hole **43***h* which penetrates the conductive part **43** in the X direction is formed in a portion at a +Z-direction end portion side of the conductive part **43**.
[0078] At both end sides in the X direction of the nozzle guard **40**, portions provided with the screw holes **43***h* of the conductive parts **43** are formed so as to be shifted from each other in the Y direction. The portion provided with the screw hole **43***h* of the conductive part **43** at the +X-direction end side is formed at a position shifted at the +Y-direction side from the Y-direction central position of the nozzle guard **40**. On the other hand, the portion provided with the screw hole **43***h* of the conductive part **43** at the −X-direction end side is formed at a position shifted at the −Y-direction side from the Y-direction central position of the nozzle guard **40**.
[0079] Referring to FIG. **7** in combination, the conductive parts **43** are fixed outside the jet modules **30**A, **30**B, and **30**C in portions extending straight from the nozzle guard **40** toward above the base member **50**. The base member **50** is provided with an opening part **52***h* through which the conductive part **43** passes. The opening part **52***h* penetrates the carriage fixation part **52** in the base member **50** in the Z direction. The opening part **52***h* is formed to have a slit-like shape extending in the Y direction along the outer circumference of the conductive part **43** when viewed from the Z direction. The conductive part **43** extends straight toward the +Z direction through the opening part **52***h* provided to the carriage fixation part **52**.
[0080] The conductive part **43** is fixed with a screw **67**. For example, the screw **67** is screwed into an internal thread **65** provided to a −Z-direction end portion of the stay **60** from an X-direction outer side through the screw hole **43***h* of the conductive part **43**. Thus, it is possible to fix the conductive part **43** to the stay **60** with the screw **67**.
[0081] The conductive parts **43** are fixed to the same member as a grounded portion of the drive board **101** for the jet modules **30**A, **30**B, and **30**C. The grounded portion of the drive board **101** corresponds to, for example, a portion to be the reference ground (a portion functioning as a reference surface of the potential) in the drive board **101**. The drive board **101** may be provided with patterned wiring to be the reference ground. In this case, the conductive parts **43** may be coupled to the patterned wiring provided to the drive board **101**.
[0082] Referring to FIG. **8**, in the present embodiment, since there are provided the conductive parts **43** which have conductivity, and are electrically coupled to the jet modules **30**A, **30**B, and **30**C and the nozzle guard **40** via the base member **50** to couple the nozzle guard **40** to FG, a conduction path via the base member **50** is formed.
[0083] This conduction path is formed including, for example, a path with the arrow V**1** along the conductive part **43** of the nozzle guard **40**, a path with the arrow V**2** along the grounded portion of the drive board **101**, a path with the arrow V**3** along the carriage fixation part **52** of the base member **50**, and a path (not shown) at an apparatus side passing through the carriage **23** (see FIG. **1**).

Coupling Member

[0084] FIG. **9** is a diagram illustrating a conduction path sandwiched by a cover **80** and the stay **60**.

[0085] Referring to FIG. **3**, FIG. **4**, and FIG. **9** in combination, the inkjet head **5**A is provided with the jet modules **30**A, **30**B, and **30**C which jet the ink, and coupling members **70** having conductivity for coupling the jet modules **30**A, **30**B, and **30**C to FG. At least a part of the coupling member **70** is disposed at an outer side of corresponding one of the jet modules **30**A, **30**B, and **30**C.

[0086] The coupling members **70** are fixed outside the jet modules **30**A, **30**B, and **30**C. Specifically, the coupling members **70** are each fixed at an outer side in the X direction of the outer end portion in the X direction of the jet modules **30**A, **30**B, and **30**C.

[0087] The coupling members **70** are fixed on a side surface in the longitudinal direction (the X direction) of the jet modules **30**A, **30**B, and **30**C. In the present embodiment, the coupling members **70** are each formed of a plate spring. The coupling member **70** is formed by applying bending processing to a plate member made of, for example, stainless steel.

[0088] The coupling member **70** is formed to have a shape in which a portion formed to have an L-shape viewed from the Y direction and a portion formed to have an L-shape viewed from the X direction are integrated with each other. The coupling member **70** is fixed to a +Z-direction end corner portion of a +X-direction end portion in the drive board **101** when viewed from the −Y direction with a screw or the like. The three coupling members **70** are provided so as to correspond to the respective jet modules **30**A, **30**B, and **30**C. The coupling members **70** are formed of the same configuration, and are arranged side by side in the Y direction.

[0089] Specifically, the coupling members **70** are each provided with a first extending portion **71** which extends from a fixation portion with the drive board **101** toward the outer side in the +X direction beyond the +X-direction outer end edge of the drive board **101**, and then extends in the −Z direction, a second extending portion **72** which extends in the +Y direction from the −Z-direction end portion of the first extending portion **71** to a position opposed to the +X-direction end surface of corresponding one of the jet modules **30**A, **30**B, and **30**C, and a third extending portion **73** which extends obliquely toward the +X direction and the +Z direction from the +Y-direction end portion of the second extending portion **72**, and then extends straight toward the +Z direction. Each of the coupling members **70** is configured to apply biasing force in at least the X direction between corresponding one of the jet modules **30**A, **30**B, and **30**C and a holding portion **83**.

[0090] The stay **60** is disposed outside the jet modules **30**A, **30**B, and **30**C. A part of the coupling member **70** protrudes to the outside of the stay **60**. The stay **60** is provided with an opening part **61***h* through which the coupling members **70** pass. An internal thread **66** is formed at the**-**Z direction side of the opening part **61***h* of a stay main body part **61**.

[0091] Specifically, the stay **60** is provided with the stay main body part **61** formed to have a plate shape having the thickness direction set to the X direction, and the longitudinal direction set to the Z direction, and projecting portions **62** projecting from both outer end edges in the Y direction of the stay main body part **61** toward an inner side in the X direction. The stay main body part **61** is provided with the opening part **61***h* for drawing out a part (a portion at the X-direction outer end side of the third extending portion **73**) of the coupling member **70** to the +X direction side beyond the stay main body part **61**. The opening part **61***h* is formed to have a slit-like shape penetrating the stay main body part **61** in the X direction, and extending in the Y direction. A part of each of the coupling members **70** protrudes toward the +X direction side beyond the stay main body part **61** through the opening part **61***h*. The internal threads **66** are formed as a pair at an interval in the Y direction.

[0092] The inkjet head **5**A is provided with the holding portion **83** which holds portions of the coupling members **70** protruding outside the stay **60** by clamping the portions with the stay **60**. The portion of each of the coupling members **70** protruding outside the stay **60** is clamped between the holding portion **83** and the stay **60**.

[0093] The inkjet head **5**A is provided with the cover **80** which covers the jet modules **30**A, **30**B, and **30**C and the stay **60**. The holding portion **83** forms a part of the cover **80**.
[0094] The cover **80** is formed to have a shape opening in the −Z direction, and covers the upper side of the jet modules **30**A, **30**B, and **30**C and so on from the +Z direction. Specifically, the cover **80** is provided with a cover main body part **81** formed to have a box-like shape having the longitudinal direction set to the X direction, and opening in the −Z direction, and extending portions **82** which extend in the −Z direction from both outer end portions in the X direction of the cover main body part **81**.
[0095] The cover main body part **81** is provided with through holes for exposing a port for the ink to inflow, connectors, and so on to the +Z direction, and so on. A part of the extending portion **82** is formed of the holding portion **83**. A screw hole **82***h* which penetrates the extending portion **82** in the X direction is provided to a portion at the −Z-direction end portion side of each of the extending portions **82**. The screw holes **82***h* are formed as a pair at an interval in the Y direction. The screw hole **82***h* is formed at a position corresponding to the internal thread **66** (a position overlapping the internal thread **66** when viewed from the X direction in an assembled state of the cover **80**).
[0096] The coupling members **70** are fixed with screws **68**. For example, in the state in which a part of each of the coupling members **70** protrudes outside the stay **60**, the screws **68** are screwed into the internal threads **66** of the stay main body part **61** from the outer side in the X direction through the screw holes **82***h* of the cover **80**, respectively. Thus, it is possible to fix the extending portions **82** to the stay **60** with the screws **68**, and at the same time, clamp the portion of each of the coupling members **70** protruding outside the stay **60** between the stay **60** and the holding portion **83** to hold the portion. In the present embodiment, it is possible to hold the portions of the respective coupling members **70** protruding outside the stay **60** with the holding portion **83** of the cover **80** in a lump.
[0097] Referring to FIG. **9**, in the present embodiment, since the coupling members **70** which have conductivity and are for coupling the jet modules **30**A, **30**B, and **30**C to FG are provided, the holding portion **83** for holding the portions of the coupling members **70** protruding outside the stay **60** by clamping the portions with the stay **60** is provided, and the holding portion **83** forms a part of the cover **80**, the conduction path sandwiched by the cover **80** and the stay **60** is formed.
Operation Method of Printer
[0098] A method of recording information on the recording target medium P using the printer **1** described above will be described.
[0099] As shown in FIG. **1**, when operating the printer **1**, the grit rollers **11**, **13** of the conveyance mechanisms **2**, **3** rotate to thereby convey the recording target medium P between the grit rollers **11**, **13** and the pinch rollers **12**, **14** in the +X direction. Further, at the same time as this operation, the drive motor **28** rotates the pulley **26** to run the endless belt **27**. Thus, the carriage **23** reciprocates in the Y direction while being guided by the guide rails **21**, **22**.
[0100] Meanwhile, in the inkjet heads **5**A, **5**B, the drive voltages are applied to the respective drive electrodes of the head chips. Thus, the thickness shear deformation is caused in the drive wall, and thus, the pressure wave is generated in the ink filling the ejection channel. Due to the pressure wave, the internal pressure of the ejection channel increases, and the ink is ejected through the nozzle hole. Further, by the ink landing on the recording target medium P, a variety of types of information are recorded on the recording target medium P.
Functions and Advantages
[0101] The inkjet heads **5**A, **5**B are each provided with the base member **50**, the jet modules **30**A, **30**B, and **30**C which are mounted on the base member **50** and jet the ink, the nozzle plate **35** provided with the nozzle holes for jetting the ink, the nozzle guard **40** made of metal for protecting the nozzle plate **35**, and the conductive parts **43** which have conductivity, are electrically coupled to the jet modules **30**A, **30**B, and **30**C and the nozzle guard **40** via the base member **50**, and couple

the nozzle guard **40** to FG.
[0102] According to this configuration, it is possible to short the nozzle guard **40** to make the nozzle guard **40** the same in potential as FG so as to prevent the static electricity from the medium from hitting, for example, the electrodes through the nozzle holes. Therefore, it is possible to prevent the damage of the inkjet heads **5**A, **5**B from the static electricity.
[0103] The conductive parts **43** in the present embodiment are fixed outside the jet modules **30**A, **30**B, and **30**C.
[0104] According to this configuration, the fixation work can be performed outside the jet modules **30**A, **30**B, and **30**C, which, therefore, makes a contribution to an improvement of the workability.
[0105] The conductive parts **43** in the present embodiment are fixed on the side surfaces in the longitudinal direction of the jet modules **30**A, **30**B, and **30**C.
[0106] According to this configuration, since there is no influence in the thickness direction of the inkjet heads **5**A, **5**B, density growth can be achieved even when disposing a plurality of inkjet heads **5**A, **5**B in the thickness direction.
[0107] The conductive parts **43** in the present embodiment are fixed with the screws **67**.
[0108] According to this configuration, since screw fixation is adopted, the conduction can more reliably be achieved.
[0109] The conductive parts **43** in the present embodiment are fixed to the same member as the grounded portion of the drive board **101** for the jet modules **30**A, **30**B, and **30**C.
[0110] According to this configuration, the potential can reliably be made the same as that of the drive board **101** without intervention of any other members due to the commonalization of the grounded portion. In addition, noise suppression effect is obtained.
[0111] The base member **50** in the present embodiment is provided with the opening parts **52***h* through which the conductive parts **43** pass.
[0112] According to this configuration, the jet modules **30**A, **30**B, and **30**C and the nozzle guard **40** can electrically be coupled through the opening parts **52***h* of the base member **50**, which, therefore, makes a contribution to shortening of the conduction path. Further, even when the base member **50** is partially made of a nonmetallic material, the nozzle guard **40** can be made the same in potential as FG.
[0113] At least corner portions of the nozzle guard **40** in the present embodiment have the drawn parts **45** to which drawing processing is applied.
[0114] According to this configuration, it is possible to prevent a gap from being formed in at least the corner portions of the nozzle guard **40**.
[0115] The side surface portion of the nozzle guard **40** in the present embodiment has the bent part **46** to which the bending processing is applied.
[0116] According to this configuration, even the portion which cannot be provided with enough length by the drawing processing alone can be stretched long enough by the bending processing, and therefore, it becomes possible to achieve the conduction with the side surface portion.
[0117] On the boundary between the drawn part **45** and the bent part **46** of the nozzle guard **40** in the present embodiment, there is formed the cutout **47**.
[0118] According to this configuration, it is possible to discharge the liquid which has entered the drawn part **45** through the cutouts **47** of the nozzle guard **40**. In addition, even when the nozzle guard **40** is made by the hybrid processing of the drawing processing and the bending processing, the cutouts **47** make the processing easy.
[0119] The conductive part **43** in the present embodiment is formed of the same member as, and integrally with, the side surface portion of the nozzle guard **40**.
[0120] According to this configuration, no dedicated conduction component is required, which makes a contribution to a reduction in the number of components.
[0121] The conductive parts **43** in the present embodiment are fixed outside the jet modules **30**A, **30**B, and **30**C in the portions extending straight from the nozzle guard **40** toward above the base

member **50**.

[0122] According to this configuration, it becomes possible to reduce the space in the installation surface direction compared to when disposing, for the fixation, a widened portion extending in the installation surface direction of the inkjet heads **5**A, **5**B. Further, a housing made of resin can be disposed between the nozzle guard **40** and the base member **50**, which, therefore, makes a contribution to the reduction in cost compared to when disposing a metallic housing.

[0123] The printer **1** according to the present embodiment is provided with the inkjet heads **5**A, **5**B described above.

[0124] According to this configuration, it is possible to obtain the printer **1** capable of preventing the damage of the inkjet heads **5**A, **5**B from the static electricity.

[0125] Incidentally, when a cover component for protecting the nozzle plate is made of metal having conductivity, and is not electrically coupled to other components, it is preferable for the cover component to be made the same in potential as FG on the following grounds.

[0126] (1) The nozzle guard is shorted to be grounded so that the static electricity from the medium does not hit the electrodes (e.g., a common electrode) and so on through the nozzle holes.

[0127] (2) A return current path (a path from the nozzle guard to the drive board) of the noise generated during driving is made the shortest so that the noise does not have a harmful influence.

[0128] (3) In order to prevent metal corrosion due to a potential difference between the nozzle guard and the electrodes (e.g., the common electrode) when these are coupled to each other via the ink, the nozzle guard is prevented from electrically floating.

[0129] In the present embodiment described above, there are provided the conductive parts **43** which are electrically coupled to the jet modules **30**A, **30**B, and **30**C and the nozzle guard **40** via the base member **50**, and are for coupling the nozzle guard **40** to FG, and the conductive parts **43** are fixed outside the jet modules **30**A, **30**B, and **30**C in the portions extending straight from the nozzle guard **40** toward above the base member **50**. Thus, (A) it is possible to prevent the damage of the inkjet heads **5**A, **5**B from the static electricity, (B) it is possible to reduce the influence of the noise during printing, and (C) it is possible to prevent the metal corrosion of the nozzle guard **40**. Further, (D) the reduction in space in the installation surface direction becomes possible, and thus, the size of the carriage **23** can be reduced, which makes a contribution to a reduction in size of the apparatus, and a reduction in cost of the conveyance system. Further, (E) it becomes possible to achieve the advantages (A) to (D) described above while keeping the strength of the inkjet heads **5**A, **5**B.

[0130] The inkjet heads **5**A, **5**B according to the present embodiment are each provided with the jet modules **30**A, **30**B, and **30**C which jet the ink, and the coupling members **70** having conductivity for coupling the jet modules **30**A, **30**B, and **30**C to FG. At least a part of the coupling member **70** is disposed at an outer side of corresponding one of the jet modules **30**A, **30**B, and **30**C.

[0131] For example, when the whole of the coupling member for coupling the jet module to FG is disposed inside the jet module, it becomes difficult to visually confirm the FG coupling at the time of assembling. In contrast, according to this configuration, since at least a part of the coupling member **70** is disposed at the outer side of the jet modules **30**A, **30**B, and **30**C, it is easy to visually confirm the FG coupling at the time of assembling. Further, it is easy to confirm the conduction even after assembling.

[0132] The coupling members **70** in the present embodiment are fixed outside the jet modules **30**A, **30**B, and **30**C.

[0133] According to this configuration, the fixation work can be performed outside the jet modules **30**A, **30**B, and **30**C, which, therefore, makes a contribution to an improvement of the workability.

[0134] The coupling members **70** in the present embodiment are fixed on the side surfaces in the longitudinal direction of the jet modules **30**A, **30**B, and **30**C.

[0135] According to this configuration, since there is no influence in the thickness direction of the jet modules **30**A, **30**B, and **30**C, density growth can be achieved even when disposing a plurality of

jet modules **30**A, **30**B, and **30**C in the thickness direction.
[0136] The coupling members **70** in the present embodiment are fixed with the screws **68**.
[0137] According to this configuration, since screw fixation is adopted, the conduction can more reliably be achieved.
[0138] The inkjet heads **5**A, **5**B in the present embodiment are each provided with the stay **60** to be disposed outside the jet modules **30**A, **30**B, and **30**C. A part of the coupling member **70** protrudes to the outside of the stay **60**.
[0139] According to this configuration, since a part of the coupling member **70** protrudes outside the stay **60**, it is easy to visually confirm the FG coupling at the time of assembling.
[0140] The inkjet heads **5**A, **5**B according to the present embodiment are each provided with the holding portion **83** which holds the portions of the coupling members **70** protruding outside the stay **60** by clamping the portions with the stay **60**.
[0141] According to this configuration, the fixation is achieved by the clamping with the stay **60** and the holding portion **83**, which, therefore, makes a contribution to an improvement in reliability of the electrical coupling.
[0142] The stay **60** in the present embodiment is provided with the opening part **61***h* through which the coupling members **70** pass.
[0143] According to this configuration, it is possible to draw a part of the coupling member **70** out of the stay **60** through the opening part **61***h* of the stay **60**.
[0144] The inkjet heads **5**A, **5**B according to the present embodiment are each provided with the cover **80** which covers the jet modules **30**A, **30**B, and **30**C and the stay **60**. The holding portion **83** forms a part of the cover **80**.
[0145] According to this configuration, since the fixation is achieved by the clamping with a part of the cover **80** and the stay **60**, no dedicated fixation component is required, which makes a contribution to a reduction in the number of components.
[0146] The printer **1** according to the present embodiment is provided with the inkjet heads **5**A, **5**B described above.
[0147] According to this configuration, it is possible to obtain the printer **1** which is easy to visually confirm the FG coupling at the time of assembling, and is easy to confirm the conduction even after assembling.
[0148] Incidentally, in industrial-use inkjet printers, since the ink is jetted, the surface of the medium is charged, and therefore, there is a possibility that electrostatic discharge occurs in the inkjet heads. Further, when driving the actuators used in the inkjet heads, a loud drive noise may be generated in some cases. In this case, there is also a possibility that the drive noise incurs some erroneous operations of other inkjet heads inside the inkjet printer. The configurations disclosed in PTL1 and PTL2 exist in order to avoid the above.
[0149] However, in the case of the configuration in PTL1, since the contact protrusion is disposed inside the head cover, it is difficult to visually confirm whether the contact protrusion has contact with the nozzle plate when viewed from the outside of the head cover. In the case of the configuration in PTL2, it is difficult to visually confirm whether the plate spring and the contact point part are coupled with each other when viewed from the outside of the main body. Further, when the whole of a coupling member for coupling the main body to a frame ground is disposed inside the main body, it becomes difficult to visually confirm the frame ground coupling at the time of assembling. Further, when it is attempted to ensure the electrical coupling only with the elastic restorative force of the plate spring, there is a concern that the coupling becomes unstable due to the vibration at the time of transportation and the vibration of the apparatus.
[0150] In contrast to the above, in the present embodiment, there are provided the coupling members **70** which have conductivity, and are for coupling the jet modules **30**A, **30**B, and **30**C to FG, the stay **60** disposed outside the jet modules **30**A, **30**B, and **30**C, and the cover **80** which covers the jet modules **30**A, **30**B, and **30**C and the stay **60**, a part of each of the coupling members

70 protrudes outside the stay **60**, and the cover **80** is provided with the holding portion **83** which clamps the portions of the coupling members **70** protruding outside the stay **60** with the stay **60** to thereby hold the portions. Thus, (A) it is easy to visually confirm the FG coupling at the time of assembling since a part of the coupling member **70** protrudes outside the stay **60**, and (B) the part of the coupling member **70** is fixed by the clamping with the stay **60** and the holding portion **83** of the cover **80**, which, therefore, makes a contribution to the improvement of the reliability of the electrical coupling. Further, (C) it is possible to achieve robustness against external factors such as an apparatus vibration due to the fixation with the clamping described above. Further, (D) when three or more jet modules **30**A, **30**B, and **30**C are installed, it is easy to achieve the FG coupling of the jet module (e.g., the jet module **30**B) disposed at the center side among them.

Modified Examples

[0151] It should be noted that the scope of the invention is not limited to the embodiments described above, but a variety of modifications can be applied within the scope or the spirit of the invention.

[0152] In the embodiment described above, there is explained the configuration in which the conductive parts are fixed outside the jet modules, but this configuration is not a limitation. For example, the conductive parts may be fixed inside the jet modules.

[0153] In the embodiment described above, there is explained the configuration in which the conductive parts are fixed on the side surface in the longitudinal direction of the jet module, but this configuration is not a limitation. For example, the conductive parts may be fixed on the side surface in the thickness direction of the jet module.

[0154] In the embodiment described above, there is explained the configuration in which the conductive parts are fixed with the screws, but this configuration is not a limitation. For example, the conductive parts may be fixed via a biasing member such as a spring, or may be fixed with an electrically-conductive adhesive or the like.

[0155] In the embodiment described above, there is explained the configuration in which the conductive parts are fixed to the same member as the grounded portion of the drive board for the jet module, but this configuration is not a limitation. For example, the conductive parts may be fixed to a different member (via a different member) from the grounded portion of the drive board for the jet module. The fixation configuration of the conductive parts can be changed in accordance with the design specification.

[0156] In the embodiment described above, there is explained the configuration in which the base member is provided with the opening part through which the conductive parts pass, but this configuration is not a limitation. For example, there may be adopted a configuration in which the base member is not provided with the opening part, and the conductive parts extend so as to bypass the base member to electrically couple the jet modules and the nozzle guard. The formation configuration of the opening part in the base member and/or the configuration of the extension of the conductive parts can be changed in accordance with the design specification.

[0157] In the embodiment described above, there is explained the configuration in which at least the corner portions of the nozzle guard have the drawn parts to which the drawing processing is applied, but this configuration is not a limitation. For example, the corner portions of the nozzle guard may have the bent parts to which the bending processing is applied.

[0158] In the embodiment described above, there is explained the configuration in which the side surface portion of the nozzle guard has the bent part to which the bending processing is applied, but this configuration is not a limitation. For example, the side surface portion of the nozzle guard may have a drawn part to which the drawing processing is applied. The configuration of the drawn part and/or the bent part can be changed in accordance with the design specification.

[0159] In the embodiment described above, there is explained the configuration in which the cutout is formed on the boundary between the drawn part and the bent part of the nozzle guard, but this configuration is not a limitation. For example, on the boundary between the drawn part and the bent part of the nozzle guard, there may be disposed a buried part for filling a gap on the boundary. The formation configuration of the cutout can be changed in accordance with the design specification.

[0160] In the embodiment described above, there is explained the configuration in which the conductive part is formed of the same member as, and integrally with, the side surface portion of the nozzle guard, but this configuration is not a limitation. For example, the conductive part may be formed of a different member from the side surface portion of the nozzle guard, and may be integrally coupled to the side surface portion of the nozzle guard. For example, the conductive part may be formed of the same member as, and integrally with, the side surface portion of the stay. The formation configuration of the conductive parts can be changed in accordance with the design specification.

[0161] In the embodiment described above, there is explained the configuration in which the conductive parts are fixed outside the jet modules in the portion extending straight from the nozzle guard toward above the base member, but this configuration is not a limitation. For example, the widened portion extending in the installation surface direction of the inkjet heads may be disposed for the fixation.

[0162] In the embodiment described above, there is explained the configuration in which the coupling members are fixed outside the jet modules, but this configuration is not a limitation. For example, the coupling members may be fixed inside the jet modules.

[0163] In the embodiment described above, there is explained the configuration in which the coupling members are fixed on the side surface in the longitudinal direction of the jet module, but this configuration is not a limitation. For example, the coupling members may be fixed on the side surface in the thickness direction of the jet module.

[0164] In the embodiment described above, there is explained the configuration in which the coupling members are fixed with the screws, but this configuration is not a limitation. For example, the coupling members may be fixed via a biasing member such as a spring, or may be fixed with an electrically-conductive adhesive or the like.

[0165] In the embodiment described above, there is explained the configuration in which the inkjet head is provided with the stay disposed outside the jet module, and a part of the coupling member protrudes outside the stay, but this configuration is not a limitation. For example, the whole of the coupling members may be disposed inside the stay. For example, the stay is not required to be disposed outside the jet modules. The installation configuration of the stay and/or the arrangement configuration of the coupling members can be changed in accordance with the design specification.

[0166] In the embodiment described above, there is explained the configuration in which the inkjet head is provided with the holding portion which holds the portions of the coupling members protruding outside the stay by clamping the portions with the stay, but this configuration is not a limitation. For example, the portions of the coupling members protruding outside the stay may be fixed only to the stay. For example, the holding portion for holding the portions of the coupling members protruding outside the stay is not required to be provided. The fixation configuration of the coupling members and/or the installation configuration of the holding portion can be changed in accordance with the design specification.

[0167] In the embodiment described above, there is explained the configuration in which the stay is provided with the opening part through which the coupling members pass, but this configuration is not a limitation. For example, there may be adopted a configuration in which the stay is not provided with the opening part, and a part of the coupling member extends so as to bypass the stay to go outside the stay. The formation configuration of the opening part in the stay and/or the configuration of the extension of the coupling members can be changed in accordance with the design specification.

[0168] In the embodiment described above, there is explained the configuration in which the inkjet head is provided with the cover which covers the jet modules and the stay, and the holding portion forms a part of the cover, but this configuration is not a limitation. For example, the holding portion may be formed of a separate member from the cover. For example, the cover which covers the jet modules and the stay is not required to be disposed. The installation configuration of the cover and/or the configuration of the holding portion can be changed in accordance with the design specification.

Second Embodiment

[0169] FIG. **10** is a partially exploded perspective view of the inkjet head. FIG. **11** is a perspective view showing an attachment portion of a stay **260** and a plate member **290**. FIG. **12** is a diagram illustrating a conduction path sandwiched by the plate member **290** and the stay **260**.

[0170] In the first embodiment described above, the description is presented citing the example in which the conduction path sandwiched by the cover **80** and the stay **60** is formed, but this example is not a limitation. For example, as shown in FIG. **12**, the conduction path sandwiched by the plate member **290** and the stay **260** may be formed. In the second embodiment, substantially the same constituents as those of the embodiment described above are denoted by the same reference symbols, and the detailed description thereof will be omitted.

[0171] Referring to FIG. **10** to FIG. **12** in combination, the stay **260** is disposed outside the jet modules. A part of the coupling member **70** protrudes to the outside of the stay **260**. The stay **260** is provided with opening parts **261***h* through which the coupling members **70** pass.

[0172] The stay **260** is provided with a stay main body part **261** formed to have a plate shape having the thickness direction set to the X direction, and the longitudinal direction set to the Z direction, and projecting portions **262** projecting from both outer end edges in the Y direction of the stay main body part **261** toward an inner side in the X direction. The stay main body part **261** is provided with the opening parts **261***h* each for drawing out a part (a portion at the X-direction outer end side of the third extending portion **73**) of each of the coupling members **70** to the +X direction side beyond the stay main body part **261**. Each of the opening parts **261***h* is formed to penetrate the stay main body part **261** in the X direction, and have a rectangular shape having rounded corners when viewed from the X direction. A part of each of the coupling members **70** protrudes toward the +X direction side beyond the stay main body part **261** through corresponding one of the opening parts **261***h*.

[0173] The inkjet head is provided with holding portions **293** which hold respective portions of the coupling members **70** protruding outside the stay **260** by clamping the portions with the stay **260**. The portion of each of the coupling members **70** protruding outside the stay **260** is clamped between the holding portion **293** and the stay **260**.

[0174] The inkjet head is provided with the plate members **290** each having the holding portion **293**. The holding portion **293** forms a part of the plate member **290**. The holding portion **293** is formed in a portion at the +Z-direction end side of the plate member **290**.

[0175] The three plate members **290** are disposed at intervals in the Y direction so as to correspond to the respective coupling members **70**. The plate members **290** are each formed to have an L-shape having the longitudinal direction set to the Z direction. Specifically, the plate members **290** are each provided with a plate main body part **291** formed to have a plate shape having the thickness direction set to the X direction, and the longitudinal direction set to the Z direction, and a projecting portion **292** projecting from the +Z-direction end portion of the plate main body part **291** toward the −X direction.

[0176] A part of the plate main body part **291** is formed of the holding portion **293**. The holding portion **293** is formed in a portion at the +Z-direction end side of the plate main body part **291**. A screw hole **291***h* which penetrates the plate main body part **291** in the X direction is provided to a portion at the −Z-direction end portion side of the plate main body part **291**. The portion provided with the screw hole **291***h* is widened in the Y direction compared to a portion at the +Z direction side of the plate main body part **291**.

[0177] The stay main body part **261** is provided with through holes **261***i* in a region at the +Z direction side of the respective opening parts **261***h*, wherein the through holes **261***i* each allows the projecting portion **292** of the plate member **290** to protrude toward the −X direction side beyond the stay main body part **261**. Each of the through holes **261***i* is formed to penetrate the stay main body part **261** in the X direction, and have a rectangular shape having rounded corners when viewed from the X direction. Each of the through holes **261***i* is formed smaller in size than each of the opening parts **261***h*.

[0178] Internal threads **266** are formed at the −Z direction side of the respective opening parts **261***h* of the stay **260**. The internal thread **266** is arranged on a Z direction line passing through the Y-direction central position of each of the opening parts **261***h*, and the three internal threads **266** are formed at intervals in the Y direction.

[0179] The coupling members **70** are fixed with screws **268**. For example, in the state in which a part of each of the coupling members **70** protrudes outside the stay **260**, the screws **268** are screwed into the internal threads **266** of the stay **260** from the outer side in the X direction through the screw holes **291***h* of the plate main body parts **291**, respectively. Thus, it is possible to fix the plate main body parts **291** to the stay **260** with the screws **268**, and at the same time, clamp the portion of each of the coupling members **70** protruding outside the stay **260** between the stay **260** and the holding portion **293** to hold the portion. In the present embodiment, it is possible to individually hold the portions of the respective coupling members **70** protruding outside the stay **260** with the holding portions **293** of the respective plate members **290**.

[0180] Referring to FIG. **12**, in the present embodiment, since the coupling members **70** which have conductivity and are for coupling the jet modules to FG are provided, the holding portions **293** for holding the portions of the coupling members **70** protruding outside the stay **260** by clamping the portions with the stay **260** are provided, and the holding portions **293** each form a part of the plate member **290**, the conduction path sandwiched by the plate member **290** and the stay **260** is formed.

[0181] The inkjet head according to the present embodiment is provided with the plate members **290** each having the holding portion **293**.

[0182] According to this configuration, it is possible to more reliably achieve the conduction compared to when fixing the coupling members **70** directly with the screws.

Other Modified Examples

[0183] It should be noted that the scope of the present disclosure is not limited to the embodiments described above, but a variety of modifications can be applied within the scope or the spirit of the present disclosure.

[0184] For example, in the embodiment described above, the description is presented citing the inkjet printer as an example of the liquid jet recording apparatus, but the liquid jet recording apparatus is not limited to the printer. For example, the liquid jet recording apparatus can be a facsimile machine, an on-demand printing machine, and so on.

[0185] In the embodiments described above, the description is presented citing the configuration (a so-called shuttle machine) in which the inkjet heads move with respect to the recording target medium when performing printing as an example, but this configuration is not a limitation. The configuration related to the present disclosure can be adopted as the configuration (a so-called stationary head machine) in which the recording target medium is moved with respect to the inkjet heads in the state in which the inkjet heads are fixed.

[0186] In the embodiment described above, there is described when the recording target medium is paper, but this configuration is not a limitation. The recording target medium is not limited to paper, but can also be a metal material or a resin material, and can also be food or the like.

[0187] In the embodiments described above, there is explained the configuration in which the liquid jet heads are installed in the liquid jet recording apparatus, but this configuration is not a limitation. Specifically, the liquid to be jetted from the liquid jet heads is not limited to what is landed on the recording target medium, but can also be, for example, a medical solution to be blended during a dispensing process, a food additive such as seasoning or a spice to be added to

food, or fragrance to be sprayed in the air.
[0188] In the embodiment described above, there is described the configuration in which the Z direction coincides with the gravitational direction, but this configuration is not a limitation. For example, the Z direction can be made coincide with a horizontal direction.
[0189] In the embodiments described above, there is explained the configuration in which the three jet modules are mounted on the base member, but this configuration is not a limitation. The number of the jet modules mounted on the base member may also be one, two, or a plural number equal to or greater than four.
[0190] In the embodiments described above, the jet modules (the head chips) of an edge shoot type are described, but this is not a limitation. For example, it is also possible to apply the invention to a head chip of a so-called side shoot type for ejecting the ink from a central part in the extending direction in the ejection channel.
[0191] Further, it is also possible to apply the invention to a head chip of a so-called roof shoot type in which the direction of the pressure applied to the ink and the ejection direction of the ink are made to coincide with each other.
[0192] In the embodiments described above, there is explained the configuration in which the inkjet head is provided with the base member, the jet modules which are mounted on the base member, and jet the liquid, the nozzle plate provided with the nozzle holes for jetting the liquid, the nozzle guard which is made of metal and protects the nozzle plate, and the conductive parts which have conductivity, are electrically coupled to the jet modules and the nozzle guard via the base member, and couple the nozzle guard to FG, but this configuration is not a limitation. For example, the nozzle guard made of metal for protecting the nozzle plate is not required to be provided. For example, the conductive parts having conductivity for coupling the nozzle guard to FG are not required to be provided. The installation configuration of the nozzle guard and/or the conductive parts can be changed in accordance with the design specification. That is, it is sufficient that the liquid jet head according to an aspect of the present disclosure is provided with the jet module configured to jet the liquid, and a coupling member having conductivity for coupling the jet module to the frame ground, wherein at least a part of the coupling member is disposed outside the jet module.
[0193] Besides the above, it is arbitrarily possible to replace the constituents in the embodiment described above with known constituents within the scope or the spirit of the invention, and it is also possible to arbitrarily combine the modified examples described above.
[0194] Although the preferred embodiments of the present disclosure and the modified examples are hereinabove described, it should be understood that these are illustrative descriptions of the present disclosure, and should not be considered as a limitation. Modification such as addition, omission, and displacement can be implemented within the scope or the spirit of the present disclosure. Therefore, the present disclosure should not be assumed to be limited by the above description, but is limited by the appended claims.

Claims

1. A liquid jet head comprising: a jet module configured to jet a liquid; and a coupling member having conductivity and configured to couple the jet module to a frame ground, wherein at least a part of the coupling member is disposed outside the jet module.
2. The liquid jet head according to claim 1, wherein the coupling member is fixed outside the jet module.
3. The liquid jet head according to claim 2, wherein the coupling member is fixed on a side surface in a longitudinal direction of the jet module.
4. The liquid jet head according to claim 1, wherein the coupling member is fixed with a screw.
5. The liquid jet head according to claim 1, further comprising: a stay disposed outside the jet

module, wherein a part of the coupling member protrudes outside the stay.
6. The liquid jet head according to claim 5, further comprising: a holding portion configured to clamp the part of the coupling member protruding outside the stay with the stay to hold the part.
7. The liquid jet head according to claim 5, wherein the stay is provided with an opening part through which the coupling member passes.
8. The liquid jet head according to claim 6, further comprising: a cover configured to cover the jet module and the stay, wherein the holding portion forms a part of the cover.
9. The liquid jet head according to claim 6. further comprising: a plate member having the holding portion.
10. A liquid jet recording apparatus comprising: the liquid jet head according to claim 1.